

Introduction to CSS

Adding interest and sparkle to web pages.



Living in Nicollet

Nicollet offers the best of both worlds – small town living and quick access to quality medical care, national retailers and other specialized services. For families working in more than one of the surrounding larger cities, Nicollet's central location offers short drive times, affordable housing and a high quality of life.

Nicollet's core residential area of well-maintained older homes is expanding with new housing developments and available building sites on all sides of town. There are also several multi-unit apartment buildings along with a mobile home park.

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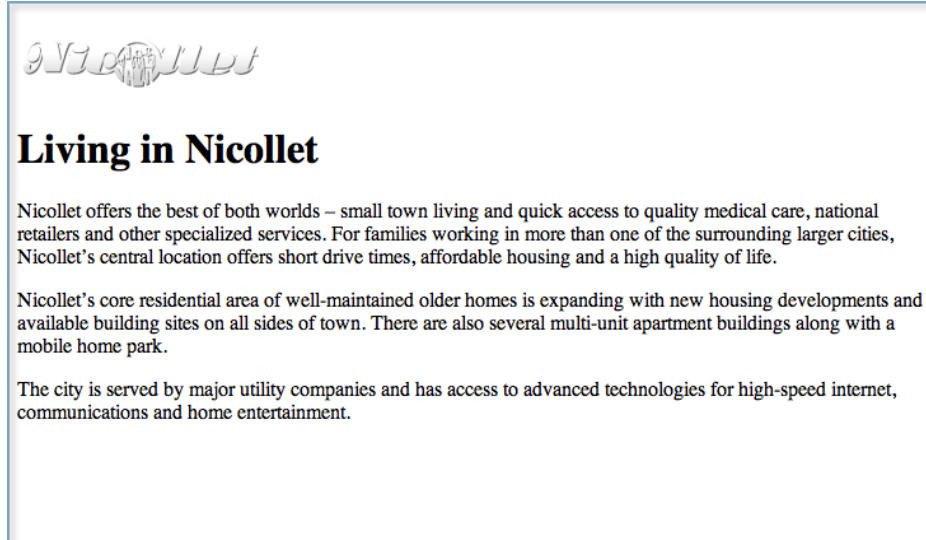


By Peter K. Johnson

Why CSS?

CSS is the frosting on the HTML cake.

With some simple lines of text you can transform a page from this:



to this:



<style> element

Start with the **<style> element**.

This tells the browser to start thinking in CSS language instead of HTML.

This element normally goes inside the **<head>** element. This puts the CSS code up in the browsers memory so it can be accessed as the browser parses through the page of HTML code.

<style type="text/css">

</style>

```
8 <head>
9   <meta charset="utf-8" />
10  <title>Nicollet County</title>
11  <style type="text/css">
12    /* The CSS for the entire page goes here */
13
14  </style>
15 </head>
16
```

Comments are marked using the **/*** and ending with the ***/**. Be nice to the browser and treat the comment markers like a word by putting a space between each marker and the comment text itself.

Anatomy of CSS

Parts of a CSS Statement

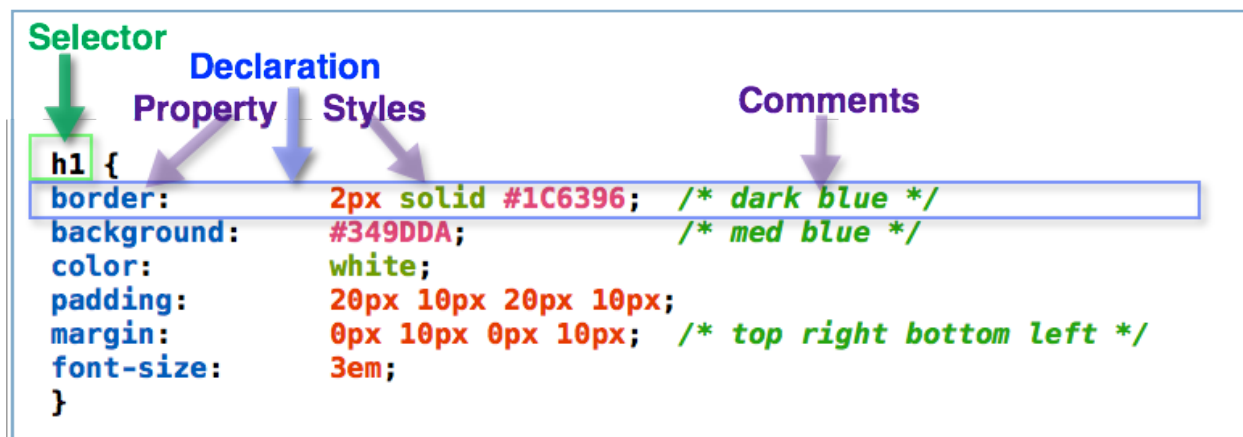
CSS text is made up of two main parts: The selector and the declarations that go with that selector.

Each declaration has a Property and a Style or value for that property. You can also add in optional comments. These comments are studiously ignored by every browser but beloved by all humans.

Keep all the selectors, properties and styles lower case except for hex color codes which are normally typed as upper case.

It doesn't matter what order the declarations are in. Often times people list the properties in alphabetical order to help them stay organized.

When multiple styles are included they can usually be in any order as well. However a few styles, such as those for margin: and padding:



CSS Punctuation

CSS is fussy. Only one character will keep it from working.

Pay attention to the punctuation because it is critical.

- Curly braces `{ }` are used to "frame" all the declarations for each selector.
- After each property, you must use a colon `:`.

- And, at the end of each style, you must use a semi-colon ";".
- Comments are marked using /* and */ and show up green in most text editors with color coding.

CSS Punctuation

```
h1 {  
border:      2px solid #1C6396; /* dark blue */  
background:  #349DDA;          /* med blue */  
color:       white;  
padding:     20px 10px 20px 10px;  
margin:      0px 10px 0px 10px; /* top right bottom left */  
font-size:   3em;  
}
```

Curly braces surround the declarations for each selector

colons mark end of each property **semi-colons mark end of each line**

/* comment markers -- ignored by the browser */

The diagram shows a CSS rule for an h1 selector. The code is color-coded: selectors are blue, property names are blue, values are red, and comments are green. Arrows point from text labels to specific punctuation marks in the code: a blue arrow points from 'Curly braces surround the declarations for each selector' to the opening curly brace; a grey arrow points from 'colons mark end of each property' to the colon after 'border'; a purple arrow points from 'semi-colons mark end of each line' to the semi-colon after 'font-size'; and a green arrow points from '/* comment markers -- ignored by the browser */' to the comment markers on the 'margin' line.

Setting up the Color Scheme

One of the most helpful tips when working with CSS is to set up a color scheme first. This will give your pages a nice consistency and make them look much more professional than picking colors willy-nilly from a color chart.

For our site, the Chamber of Commerce for Nicollet, Minnesota we want a professional look to match their existing site using blues and grays with a contrasting light yellow.

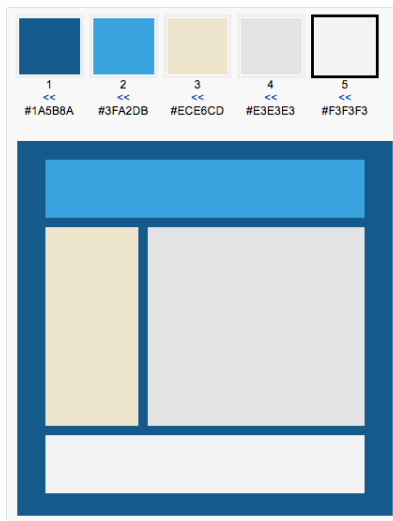
#1A5B8A - Dark Blue

#3FA2DB - Medium Blue

#ECE6CD - Light Cream

#E3E3E3 - Light Gray

#F3F3F3 - Off white



Here are [several color scheme generators](#) that you can use to build your color scheme. Or do a web search for: HTML color scheme generator.

Once you have the color scheme established, print it out and keep it handy as you develop the CSS.

Change the background

You can use CSS to style any HTML element. Let's start with the <h1> element on the page. Change the background of this block element.

Here is the code:

```
h1 {
background:           #349DDA;    /* medium blue */
}
```

```
11  <style type="text/css">
12  h1 {
13  background:           #349DDA;    /* medium blue */
14  }
15  </style>
```

And here are the results:



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Change the foreground

You can also change the color of the text. Add a rule to the h1 selector to make the text white.

Inside the h1 selector:

color: **white;**

```

11 <style type="text/css">
12 h1 {
13   background: #349DDA; /* medium blue */
14   color: white;
15 }
16 </style>

```

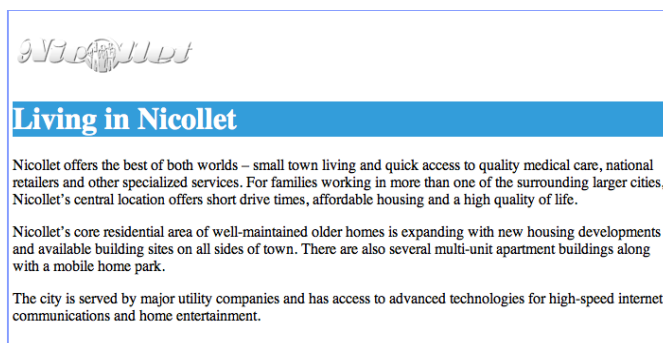
Notice how the CSS code is typed in three separate columns: Property Style Comment

This will make your CSS much easier to read and maintain.

Also, notice that common color names can be used. When using hex codes include the hash mark "#" but leave it off if you use a color name. This is a great way to add color when you don't need the precision of RGB hexadecimal codes.

All the common color names work including: white, black, pink, green, purple, blue, and yellow. Here is a [list of color names supported by all browsers](#).

Here is what the page looks like now:



Add padding around the text

You probably noticed that the text of the h1 is almost touching the edges of the box. This wasn't noticeable until we changed the background color of the box.

By adding padding to the h1 selector we can add in a little more space around the text:

padding: 20px 10px 20px 10px;

The px stands for pixels.

```

16 h1 {
17   background: #349DDA;           /* medium blue */
18   color: white;
19   padding: 20px 10px 20px 10px; /* top right bottom left */
20 }
```

If the padding is the same all around you only need a single number.

padding: 20px; /* same padding all around */

If the padding is the same for the top/bottom and different for left/right then you can use two numbers:

**padding 20px 10px; /* 20px on top and bottom
with 10px on left and right */**

Using four numbers you can customize all four sides. Think of this as a clock starting at midnight and moving around clock-wise: top, right, bottom, left.

**padding 20px 10px 5px 30px; /* top right
bottom left */**

Here are the results:



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Add margin around the block

You can also put space around an object itself, not just the text.

Add a margin of 10px to either side of the h1 selector:

margin: **0px 10px; /* top/bottom right/left */**

Notice how you can alphabetize the declarations.

```

16  h1 {
17  background: #349DDA;           /* medium blue */
18  color:      white;
19  margin:     0px 10px;           /* top/bottom left/right */
20  padding:    20px 10px 20px 10px; /* top right bottom left */
21  }
22

```

Here are the results:



Add a border

You can put a border around any object on the page. This is especially useful when you need to see where various objects on the page start and end.

For now, put a border around the h1 to have it stand out on the page a bit more.

If you like to type, you can use three different rules like this:

```
border-width:      1px;
border-color:      #1C6396;      /* dark blue */
border-style:      solid;
```

Or you can simply combine them as one, using the border property. The three styles can be listed in any order.

```
border:            1px solid #1C6396;      /* dark blue */
```

```
16  h1 {
17    background: #349DDA;      /* medium blue */
18    border: 1px solid #1C6396; /* dark blue */
19    color: white;
20    margin: 0px 10px;        /* top/bottom left/right */
21    padding: 20px 10px 20px 10px; /* top right bottom left */
22  }
23
```

Here are the results. Notice how subtle the effect is.



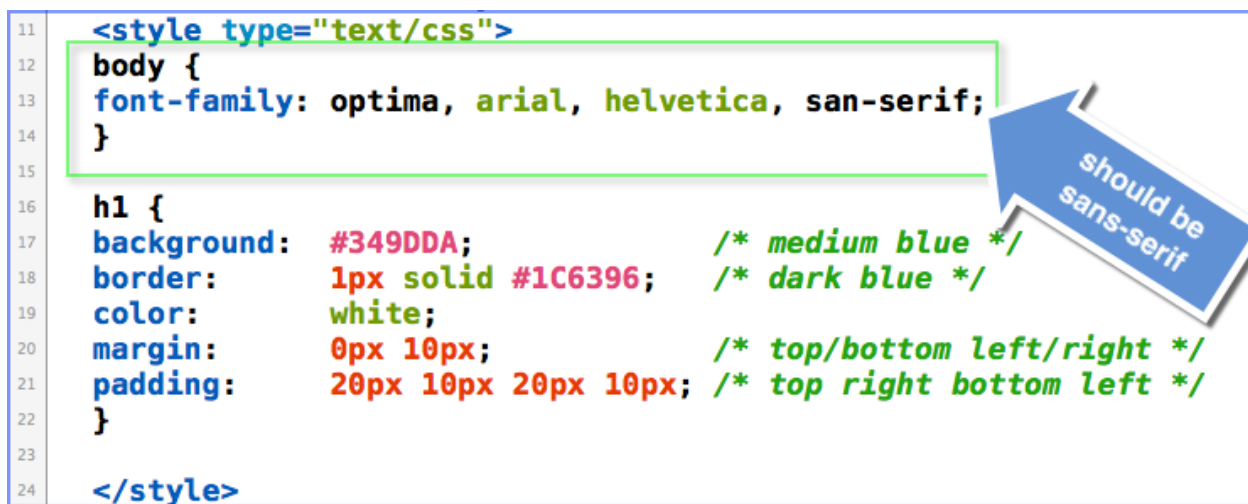
Change the font

Most browsers display text using the font Times Roman. This is not the most readable font on the web and makes pages look pretty boring.

We could add the font-family: property to the h1 selector but we want all the text on the site to be the same font to give the page a sense of unity.

To do that we will add font-family: to the body selector.

```
body {
font-family:    optima, arial, helvetica, sans-serif;
}
```



If a font has a space in the name such as "courier new", enclose it in quotation marks.

Here is what the page looks like using the Optima typeface:



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As a web developer, you don't know who is looking at the web page or what fonts they have loaded on their computer or mobile device.

By listing multiple fonts, from best to least desirable you can have more control over how the page appears on all types of systems. If the browser can't find the first font, it looks for the next one on the list until it finds a match. If it doesn't find a match it uses Times Roman.

If you want to test a font, just put an x in front of each one and display the page. That will fool the browser, forcing it to display the next font in the list. For example:

```
body {  
font-family:      Xoptima, arial, helvetica, sans-serif;  
}
```

Change the font-size

You may want to increase the size of all the fonts on the page. This is especially helpful if the majority of users are over 50 years old.

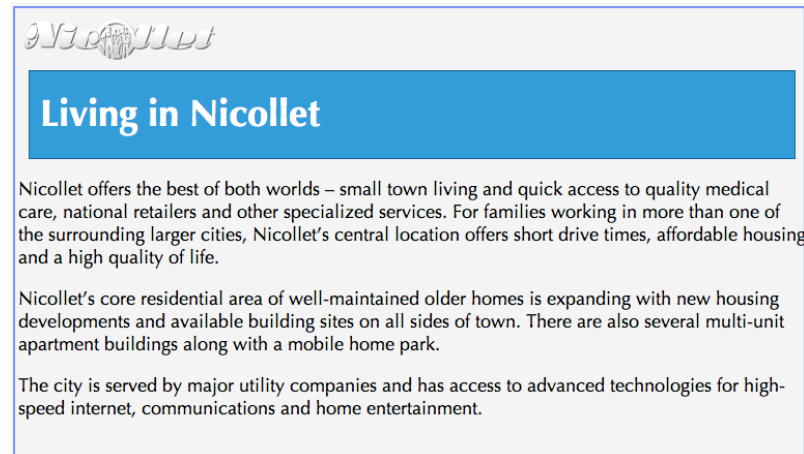
By changing the font-size in the body selector you can have all the other adjust automatically. Let's soften up the background a bit by using the light gray from our color scheme. Again, this is very subtle, but notice how it makes the h1 text (which is white) stand out a little more.

The properties were added in alphabetical order inside the body selector.

```
background:                #F4F4F4;        /* light
gray */
font-family:                optima, arial,
helvetica, sans-serif;
font-size:                  18px;
```

```
12  body {
13  background:    #F4F4F4;        /* light gray */
14  font-family:  optima, arial, helvetica, san-serif;
15  font-size:    18px;
16  }
17
```

Here are the results:



Nicollet

Living in Nicollet

Nicollet offers the best of both worlds – small town living and quick access to quality medical care, national retailers and other specialized services. For families working in more than one of the surrounding larger cities, Nicollet's central location offers short drive times, affordable housing and a high quality of life.

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Styling other elements - img

You can use CSS to style any of the HTML elements.

If you change the float: property for the img selector you can have the logo line up to the left of the h1 text. We can also add padding on the left and right side of the image as well as a margin on the top of the image.

```
img {  
    float:          left;  
    padding:        0px 20px;  
    margin:         1em 0 0 0;  
}
```



```
26  img {  
27    float:          left;  
28    padding:        0px 20px;  
29    margin:         1em 0 0 0;  
30  }
```

float: tells the browser to align the selector to the left or right. Any other elements that will fit can "float" into space on the right.

padding: tells the browser to add extra space between the image and surrounding objects. In this case, we added 20px on the left and right and made sure there was nothing added to the top and bottom.

Using em measurements for dynamic sizing

margin: tells the browser to add extra space outside the image. In this case, we added 1em. An em is the size of a capital letter "M" in the body of the page. We already set the font-size in the body selector to 18px so this will add the height of an 18px "M" along the top of the image.

The value of using an em measurement is that every em will automatically adjust whenever the font-size in the body selector is changed. This allows the

web page to be fluid because everything will change in size dynamically. Perfect when displaying the page from a desktop browser to the tiny screen of a mobile phone!

Here is what the results look like. (Don't worry, we will adjust the h1 selector so the logo and text line up on the next page.)



Living in Nicollet

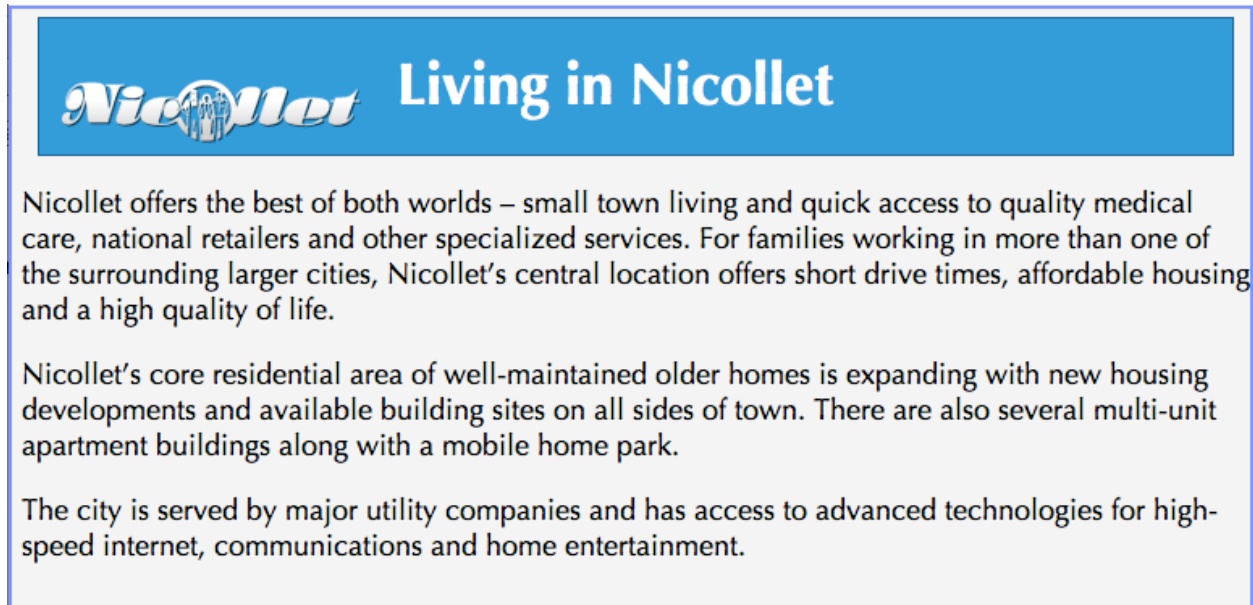
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Noodling Around

We have the logo floating to the left of the h1 text. But we want them to line up.



This is where noodling comes in handy. (That's a technical term used by programmers across the universe.)

Inside the h1 selector change the font-size using em spacing. At the same time put a border around the img selector so you can see where it is floating.

In the h1 selector add:

```
font-size:      3em;      /* 3x the height of the  
capital letter "M" */
```

In the img selector add:

```
border:      2px purple solid;  /* for debugging */
```

```

18 h1 {
19   background: #349DDA;           /* medium blue */
20   border: 1px solid #1C6396;    /* dark blue */
21   color: white;
22   font-size: 3em;
23   margin: 0px 10px;             /* top/bottom left/right */
24   padding: 20px 10px 20px 10px; /* top right bottom left */
25 }
26
27 img {
28   border: 2px purple solid;      /* for debugging */
29   float: left;
30   padding: 0px 20px;
31   margin: 1em 0 0 0;            /* 1 "M" margin on top and bottom */
32 }
33

```

The font-size for the h1 is 3em or 3 x the height of the letter "M" in the body. The border for the img is set to 2px and purple so it really stands out.

You can either noodle around with the margin in the img selector as well as the font-size in the h1 selector. By working with these two you can get the logo aligned with the text. Also, you can bet a better balance between the two objects by making the text larger.

Here are the results:



You can simply delete the border: in the img selector or comment it out to

see the finished heading.



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Using `<span>`

So far we have been using the standard HTML selectors: `body`, `h1`, and `img`.

But, you can make your own selectors. These are normally a `<div>` element or a `<span>` element.

`<div>` elements will mark out a block object while `<span>` is used to designate an in-line object.

As an example, the client wants to highlight the first few words of each paragraph. She wants the text larger and a different color.

First, we have to go into the HTML code and mark those phrases. We will use the `<span>` element because we don't want the text to make a new block (like the `h1` automatically does.) We want it to remain part of the paragraph.

Right after each `<p>` tag add a `<span>` and `</span>` around the phrase to be highlighted. Name each `<span>` the same using the `class=""` attribute. This will put all the spans in the same class or group so you only need one CSS selector.

```

36 <body>
37   
38   <h1>Living in Nicollet</h1>
39
40   <p>
41     <span class="highlight">Nicollet offers the best of both
      worlds</span> – small town living and quick access to quality
      medical care, national retailers and other specialized
      services. For families working in more than one of the
      surrounding larger cities, Nicollet's central location offers
      short drive times, affordable housing and a high quality of
      life.
42   </p>
43
44   <p>
45     <span class="highlight">Nicollet's core residential area</span>
      of well-maintained older homes is expanding with new housing
      developments and available building sites on all sides of town.
      There are also several multi-unit apartment buildings along
      with a mobile home park.
46   </p>
47
48   <p>
49     <span class="highlight">The city is served by major utility
      companies</span> and has access to advanced technologies for
      high-speed internet, communications and home entertainment.
50   </p>
51 </body>
52 </html>

```

In the <style> element add CSS code for the highlight class. In CSS the class is designated using a period "." as the first character of the selector:

```

.highlight {
  color:          #1C6396; /* dark blue */
  font-size:      1.25em;
  font-weight:    600;
}

```

```

32
33 .highlight {
34 color:          #1C6396; /* dark blue */
35 font-size:      1.25em;
36 font-weight:    600;
37 }
38
39 </style>

```

The font-size for these spans of text will be 1 1/4 x the size of the letter "M" (or 125% larger than an "M").

The font-weight can be 0-900, the higher the number the more bold the text.

Here are the results:



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Styling the <p> elements

One of the nice things about CSS is that you can write one selector for all the objects on a page.

Let's change all the paragraphs at once by changing the space between each paragraph as well as setting the margins to match the h1 block.

```
p {
margin:          1.50em 10px 1.50em 10px;
}
```

This also could have been written as:

```
p {
margin:          1.50em 10px;
}
```

```
26
27  img {
28  float:         left;
29  padding:       0px 20px;
30  margin:        1em 0 0 0;          /* 1 "M" ma
31  bottom */
32
33  p {
34  margin:        1.50em 10px 1.50em 10px;
35  }
36
37  .highlight {
38  color:         #1C6396; /* dark blue */
39  font-size:     1.25em;
40  font-weight:   600;
41  }
42
43  </style>
```

Staying organized

A nice way to organize your CSS is to put all the basic HTML selectors at the top of the list followed by all the class entries.

Many people also put the selectors in alphabetical order so they can find a selector quickly. Often a CSS file will have several hundred lines of code.

Here are the final results. Notice how the left and right margin of the <p> match up with the left and right margin of the <h1>.



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Best Practices

(1) Write a single CSS attribute at a time and then view the results in a browser.

It only takes a single character to ruin a good CSS line of code. Check each line by viewing the page in a browser. What would you rather do, find a missing colon in one line of code or an entire page?

(2) Format your CSS code into three columns.

A browser can read CSS no matter how it is formatted as long as the right punctuation is used.

But... which format of code would you prefer to work on?

This:

```

1 body{background:#F4F4F4;font-family: optima, arial, helvetica,
   san-serif;font-size:18px;}
2
3 h1{background:#349DDA;border:1px solid #1C6396;color:
   white;font-size:3em;margin:0px 10px;padding:20px 10px 20px
   10px;}
4
5 img {float:left;padding:0px 20px;margin:1em 0 0 0;}
6
7 p {margin:1.50em 10px 1.50em 10px;}
8
9 .highlight{color:#1C6396;font-size:1.25em;font-weight: 600;}
10

```

Or this:

```

12 body {
13   background: #F4F4F4;           /* light gray */
14   font-family: optima, arial, helvetica, san-serif;
15   font-size: 18px;
16 }
17
18 h1 {
19   background: #349DDA;           /* medium blue */
20   border: 1px solid #1C6396;     /* dark blue */
21   color: white;
22   font-size: 3em;
23   margin: 0px 10px;              /* top/bottom left/right */
24   padding: 20px 10px 20px 10px; /* top right bottom left */
25 }
26
27 img {
28   float: left;
29   padding: 0px 20px;
30   margin: 1em 0 0 0;            /* 1 "M" margin on top and
   bottom */
31 }
32
33 p {
34   margin: 1.50em 10px 1.50em 10px;
35 }
36
37 .highlight {
38   color: #1C6396;               /* dark blue */
39   font-size: 1.25em;
40   font-weight: 600;
41 }

```

(3) Write to your future self.

It always happens, you get the CSS working and three months later you have to make some changes. First, you can't find the code you want, and then something you change breaks three other things. All the sudden your web page is a mess.

Be nice to yourself. Add comments explaining exactly what each line of code is doing. Block out different sections of your CSS code so you can find it quickly in the future. And if it isn't you, someone else will say "thank you" when they have to open up your pages and perform maintenance on the site.

(4) Write general and then get specific.

Use the cascading part of Cascading Style Sheets. A good example in this tutorial is the h2 header. First, a general CSS rule was written covering all the headers. Immediately following was a rule which customized the h2 header. The second set of CSS rules automatically overrode the more general block of code.

(5) Always simplify.

Don't add duplicate CSS code unless you have a specific reason.

(6) Use a CSS Validation tool

All it takes is one missing semicolon ";" or confusing a colon ":" with a semi-colon ";".

There are several tools online that will validate your CSS and help you find errors in your code. Do a web search for: CSS Validator

Here is a [CSS Validation Program the W3C has developed](#)

Debugging Your CSS

Maintain A Checklist in Your Brain

Professional coders solve problems quickly. They do this through experience and by setting up checklists in their brain.

Here's the debugging checklist that was introduced earlier in this tutorial. The next time you have a problem go through this list of questions and most likely you will find what is causing your CSS to break.

If your CSS does not display here is a checklist to go over to help you debug your code:

1. Is there a colon ":" after each property? (Make certain it isn't a semi-colon)
2. Is there a semi-colon ";" after each value?
3. Did you spell everything correctly? Look things up, don't guess! (selectors, font names, attributes)
4. Do you have an open "{" and closed curly brace "}" for each selector?
5. Are all the values valid? 000099 is not valid, it must have the # in front of it.

Change the order of things

Often a property/value pair will be wrong causing all the other properties after it to be ignored by the browser. You may be focused on a property that is correct but an earlier property is keeping it from working.

Move the property you want to test up to the top of the property list and see if it starts working. If so, check the other property/value pairs (going through the numbered checklist again) to find out which property is really broken.

Use the border: property to see what is going on

Add this one line as the first line to the selector that you are having trouble with:

border: 1px solid purple;



You can set the pixel width as wide as you want and use any color name you want. Purple, pink, red, green, and blue quick and easy to see. Using outrageous colors makes the block stand out and easy to track.

If the border doesn't display then either you are not viewing the same file that you are editing (see next tip), or you are using the wrong selector.

If you have multiple selectors try setting up just one item as the selector (after the multiple selectors, so it doesn't get overwritten).

If you have lots of CSS do a Find (CTRL F in your text editor) for the selector you are working on and see if that selector is getting reset later on in your CSS code.

If so, simplify and clean up your code!

Make certain you are viewing the same file you are editing

The best way to make certain you are viewing the same file you are editing is to make a simple change in the XHTML code. I like to add an "XXX" to the end of a <h1> or the beginning of a <p> element.

Save the file and then view the file. If the "XXX" shows up you know you are looking at the same file. If not, it's time to close out all those extra browser windows you have open (keep things simple and stay organized) and do a File/Open File selecting the .html file you are actually working on.

Summary

What have you learned?

You now have the basics of CSS in a sample file.

Here are some of the techniques that you should now be able to use on your own web pages:

1. Setting up a color scheme and quick reference page printed out.
2. Changing the foreground and background of any HTML object on the page.
3. Adding padding around text and margins around any HTML object.
4. Using borders to appearance as well as to help you determine what is happening with an HTML object.
5. Use different fonts and font sizes.
6. Using the `<span>` tag along with the `class=""` attribute to name groups of similar items.
7. Refer to a specific class in your CSS code (using the period ".")
8. Debugging tips to help you fix any CSS errors
9. How to validate your CSS code to discover errors.

Credits

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